

Lab assignment 5 (GROUP ASSIGNMENT) Complete

- Face Detection with Deep Learning

17.11.2025

Attempt 1



Review Feedback

Attempt 1 Score:

7.11.2025

Complete



Add Comment

Anonymous Grading: no

Unlimited Attempts Allowed

22.9.2025 to 17.11.2025

Details

In this lab exercise, each Student or GROUP will use the code from the Face Recognition Github repo to recognize and manipulate faces from Python or from the command line with the simple face recognition library.

!! NOTE: This does not work on the Windows Subsystem for Linux (WSL) due to issues with hardware addressing on the hypervisor. **!!**

[More information on the API calls](https://face-recognition.readthedocs.io/en/latest/face_recognition.html)  [_face_recognition.html](https://face-recognition.readthedocs.io/en/latest/face_recognition.html)

1. Use the following [GitHub repo](https://github.com/ageitgey/face_recognition)  [_face_recognition](https://github.com/ageitgey/face_recognition) to set up and run face detection API.

Locally:

Create file "doorcam.py" and use [\[this code\]](https://gist.githubusercontent.com/ageitgey/84943a12dd0d9f54e90f824b94e4c2a9/raw/906b921481a0f18ddf1fcbb1e8a45d58fbde3b5e/doorbell_camera.py)  [_doorbell_camera.py](https://gist.githubusercontent.com/ageitgey/84943a12dd0d9f54e90f824b94e4c2a9/raw/906b921481a0f18ddf1fcbb1e8a45d58fbde3b5e/doorbell_camera.py) to set up and run face detection API.

Change line "`rgb_small_frame = small_frame[:, :, :-1]`" to "`rgb_small_frame = cv2.cvtColor(small_frame, cv2.COLOR_BGR2RGB)`".

2. The first task in this lab assignment is to use the face_recognition API like in the guide above.

When everything is set up, it should be possible to run this line "`python3 doorcam.py`" and get a beautiful result like the one shown below.

Deliver screenshots like the picture below, **one for each group member**

(If you get an error: "Illegal instruction (core dumped)" or something similar, use the answer from the [stack overflow](https://stackoverflow.com/questions/65631801/illegal-instructioncore-dumped-error-on-jetson-nano)  [_illegal-instructioncore-dumped-error-on-jetson-nano](https://stackoverflow.com/questions/65631801/illegal-instructioncore-dumped-error-on-jetson-nano)) and use "`python3`" instead of "`python`")

3. The second task in this lab assignment will require the group to take a group photo and individual photos of each member of the group.

Create a folder called "**known_people**" where you put the photos from each member, and the file name is the person's name.

Create a folder called "**unknown_people**" where you put the group photo.

Now navigate to the folder where these newly created folders are located and run the following line in the terminal to use the API and print the face distance calculated for each match: "**face_recognition --show-distance true ./known_people ./unknown_people**"

This will use the known peoples' faces and write to the terminal what picture it has located a face, and then the name of the face detected.

The tolerance for the match has the default value of 0.6, this can be adjusted with "--tolerance 0.54" for a tolerance value of 0.54.

Deliver a pdf file with the output images from both folders and screenshots of the terminal output from the terminal. This must be delivered through Canvas.

The code must be uploaded to your git repository (the IKT213 assignments repository you have been using through the course).

If you need help: [Lab Information \(https://uia.instructure.com/courses/18389/pages/lab-information?wrap=1\)](https://uia.instructure.com/courses/18389/pages/lab-information?wrap=1)
